Transport Reviews

ISSN: 0144-1647 (Print) 1464-5327 (Online) Journal homepage: www.tandfonline.com/journals/ttrv20

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To cite this article: Timo Välilä (2025) The economic impact of transport infrastructure: a review of project-level vs. aggregate-level evidence, Transport Reviews, 45:4, 459-481, DOI: 10.1080/01441647.2025.2476012

To link to this article: https://doi.org/10.1080/01441647.2025.2476012

Published online: 11 Mar 2025.

TRANSPORT REVIEWS
2025, VOL. 45, NO. 4, 459–481
https://doi.org/10.1080/01441647.2025.2476012

Routledge
Taylor & Francis Group

The economic impact of transport infrastructure: a review of project-level vs. aggregate-level evidence

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ABSTRACT

This article aims to assess how well the economic impact of transport infrastructure is understood. Analyses of large samples of individual investment projects based on data in the so-called Flyvbjerg database have documented systematic construction cost overruns and first-year traffic demand shortfalls. They have interpreted these findings as being indicative of negative social welfare consequences of typical transport infrastructure investment projects, but that interpretation has been challenged on methodological grounds. The evidence base of ex post social cost–benefit analyses is growing and it, too, has challenged the suggestion that transport infrastructure projects destroy social welfare. Overall, it seems fair to conclude that our understanding of the project-level social welfare consequences of transport infrastructure is improving but remains far from conclusive. In contrast, aggregate-level quantitative meta-analyses provide robust results of a significant and positive relationship between transport infrastructure and economic activity at the level of regions or countries, especially in the long run and at higher levels of geographical aggregation. These aggregate-level results imply that project-level analyses should indeed consider the entire life cycle of projects, not just construction and the first year of operation, and that they should also acknowledge the presence of network effects as well as wider economic impacts – as difficult to measure and controversial as they are.

ARTICLE HISTORY
Received 11 August 2024
Accepted 28 February 2025

KEYWORDS
Transport infrastructure; economic impact; social welfare; output elasticity; meta-studies

JEL CLASSIFICATION CODES
R42; H43; H54; O18; C83

1. Introduction

The economic impact of transport infrastructure has been studied quite extensively at the level of individual projects and also at the aggregate level of regions and countries. Indeed, the number of studies at both project and aggregate level has been so high as to induce the production of meta-studies, which combine the findings of other studies and analyse that sample of other studies quantitatively.

The largest source of information about the performance of individual investment projects is the so-called "Flyvbjerg database" (thus called in Flyvbjerg & Gardner, 2023,

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Appendix A). The database has served mainly as a basis for research on the magnitude of and causes for construction cost overruns and, to a more limited extent, benefit (traffic demand) shortfalls in the first year of operation in individual projects. That research has famously given rise to the Iron Law of Project Management, which has it that large projects tend to be "Over time, over budget, under benefits, over and over again" (Flyvbjerg, 2017). Importantly, analyses based on the database have linked the Iron Law to negative social welfare consequences of transport infrastructure investment projects.

Ansar et al. (2016) contrast that rather gloomy project-level evidence with the more upbeat aggregate-level evidence of the relationship between public capital (infrastructure) and growth. The aggregate-level evidence considered by Ansar et al. consists of a selection of individual empirical studies on the relationship between public capital or infrastructure and growth. Based on the project-level and aggregate-level evidence reviewed, Ansar et al. (2016, pp. 361–362) summarise:

> … a salient paradox in the theory on infrastructure. The macro-level school of thought that has dominated the mainstream discourse in economics has argued that increased public-sector investment in infrastructure (particularly in transport) increases the efficiency and profitability of the business sector; [and] this increase stimulates business investment in private capital … In contrast, micro-level evidence from case studies and large datasets, typically published in planning and management journals, has shown that the financial, social, and environmental performance of infrastructure investments is, in fact, strikingly poor … How can poor outcomes of individual infrastructure investment projects amount to economic welfare in the aggregate?

Ansar et al. challenge the notion that infrastructure, especially transport infrastructure, is "good" at the aggregate level and conclude instead that the project-level evidence of poor performance is more compelling.

A re-evaluation of this "paradox" is warranted by voluminous new evidence that has emerged since the article by Ansar at al. At the level of individual projects, a small but increasing body of literature has been published that surveys samples of ex post social cost–benefit analyses (CBA) of transport infrastructure projects, instead of just forecast errors in construction costs and first year traffic demand. In addition, the link between the data on forecast errors in the Flyvbjerg database and the conclusions concerning projects' social welfare consequences based on those forecast errors has been questioned (Välilä, 2024). At the aggregate level, several quantitative meta-studies have been published to analyse the vast body of empirical literature that estimates the impact of transport infrastructure on economic activity.

Therefore, the purpose of this article is twofold. First, the conclusions of project-level analyses based on the Flyvbjerg database are assessed methodologically and contrasted with the available ex post evidence of the social welfare consequences of transport infrastructure projects. Second, the project-level evidence is compared with the aggregate-level evidence in terms of the methodologies used, measures of economic impact employed, and results obtained. The review of the project-level and aggregate-level evidence will allow us to understand what exactly both measure; whether their results do, indeed, constitute a paradox; and what can be confidently concluded about the consequences of transport infrastructure for social welfare and economic activity.

Section 2 reviews the research on the performance of transport infrastructure investment projects based on the Flyvbjerg database, and it also reviews published surveys of samples of ex post CBA of transport infrastructure projects. The focus is on the social welfare impact of transport infrastructure projects as estimated using CBA, both because analyses based on the Flyvbjerg database draw conclusions about projects' social welfare consequences and because social welfare is the preferred metric in microeconomic impact analysis. Section 3 turns to aggregate-level meta-analyses, reviewing the methodology and results of quantitative meta-studies on the relationship between transport infrastructure and economic activity. Section 4 combines and contrasts the insights from sections 2 and 3. Section 5 concludes and identifies some topics for future research.

2. Project-level evidence

2.1. The Flyvbjerg database and analyses thereof

The Flyvbjerg database contains over 16,000 projects of 25 distinct "types", including seven categories pertaining to transport infrastructure. The database includes project-level information from primary sources and from secondary sources on account of continuous and systematic tracking of other studies, and it is therefore considered to "allow meta-analysis" (Flyvbjerg & Bester, 2021, Appendix, p. 412).

The Flyvbjerg database is not accessible,[1] so this section is based on published research that draws on the database. A significant share of the research based on the Flyvbjerg database has concerned major projects, defined as projects with "costs" in excess of USD100 million (Flyvbjerg, 2009, p. 345, footnote 1), or mega-projects, with "budgets" in excess of USD1 billion (Flyvbjerg & Gardner, 2023, p. 5). However, in an early article referring to the database and focussing on 258 transport projects, Flyvbjerg et al. (2003, p. 74) reports that projects with construction costs in real (inflation-adjusted) terms ranging from USD1.5 million to USD8.5 billion were included, with the majority of the projects below USD100 million. None of the articles quoted in this section reports the results by project size, so it is not possible to identify any systematic differences between small and large projects below.

A description of data collection procedures for the Flyvbjerg database is contained in Flyvbjerg and Bester (2021, Appendix). The Flyvbjerg database contains information about forecast and actual construction costs, valued likely at market or accounting prices and measured in real terms, excluding financing costs as well as non-market costs. It also contains information about demand (traffic volumes), both forecast and actual in the first year of operation. The precise unit of measurement is not systematically disclosed; it is only defined as the unit used by project planners (Flyvbjerg & Bester, 2021).

The published articles drawing on the database report and analyse forecast errors for construction costs and first-year demand for different types of investment projects. Forecast errors are calculated either as the difference between the actual outcome and the estimate at decision-making, relative to that estimate (Flyvbjerg et al., 2002; Flyvbjerg, 2005), or as ratios of actual outcomes to forecasts at decision-making (Flyvbjerg, 2016; Flyvbjerg & Bester, 2021).

Articles ranging from Flyvbjerg et al. (2002) to Flyvbjerg and Bester (2021) discuss the limitations of the database and any analysis based thereon, and they point out that while it is "not perfect by any means" (Flyvbjerg et al., 2002, p. 295), the database is the best one available. A number of methodological controversies relevant for the database are aired in the exchange between Love and Ahiaga-Dagbui (2018) on one hand and Flyvbjerg et al. (2018) on the other hand.

It is, indeed, important to recognise that the database makes it possible to measure, analyse, and draw conclusions about forecasting errors in construction costs and first-year demand, but that any further-reaching conclusions about the performance or economic impact of investment projects are open to question. Projects generate both costs and benefits throughout their life cycles beyond construction and first-year operation, and they also impact markets beyond their own primary markets, generating indirect and wider economic impacts over their long life cycles. Given the difficulty of collecting consistent data on these impacts for large samples of projects, they are not included in the Flyvbjerg database. Against this background, the scope for drawing conclusions about the longer-term and wider economic impact of transport infrastructure investment projects based on the database is discussed in greater detail at the end of this section.

The articles that include analyses of construction cost and traffic volume performance of transport projects draw on different generations of samples from the Flyvbjerg database. A sample of 258 transport infrastructure projects is analysed in Flyvbjerg et al. (2002); Flyvbjerg et al. (2003); as well as in Flyvbjerg (2009). A different sample of 95 road (74) and rail (21) infrastructure projects in China is analysed in Ansar et al. (2016). Flyvbjerg (2016) considers an extended sample of 2,062 projects in eight different infrastructure sectors, five of which are in transport. Flyvbjerg and Bester (2021) appear to analyse the same sample. Flyvbjerg and Gardner (2023, Appendix A), refer to the entire database of 16,000 projects.

Combining the information on construction costs and traffic volumes, Flyvbjerg (2016, Table 1) as well as Flyvbjerg and Bester (2021, Table 1) calculate ratios of actual to forecast cost and traffic performance. These results are reported in Table 1, which also shows the ratio of the two other ratios in its final column.

Flyvbjerg (2016) as well as Flyvbjerg and Bester (2021) consider the ratio of actual to forecast traffic volume in the first year of operation as indicating the shortfall or excess of benefits generated by the project. It is thus implicitly assumed that the proportionality of actual to forecast traffic volume translates into actual to forecast benefits. Therefore, they consider the ratio of traffic shortfall/excess to construction cost overrun as indicative of estimation error in projects' "benefit–cost ratio". They conclude: " … a typical ex-ante benefit–cost ratio produced by conventional methods is overestimated by between approximately 50 and 200%, depending on investment type" (Flyvbjerg & Bester, 2021, p. 402). Note, indeed, that the data reported in the articles underlying Table 1 allows for conclusions about forecast errors in "benefit–cost ratios", but not about their actual or forecast levels.

The documented performance in terms of construction costs and traffic volume in the first year of operation is linked to the economic impact of projects in number of articles. Flyvbjerg et al. (2002, pp. 290–291) conclude that the underestimation of construction costs is "likely to lead to the misallocation of scarce resources", while also pointing out that all types of infrastructure projects, not just transport, are susceptible to such underestimation. Flyvbjerg et al. (2003, p. 86), in turn, suggest that decision-making on infrastructure projects ignores or belittles the underestimation of construction costs, "to the detriment of social and economic welfare". Flyvbjerg (2009, p. 353) asserts that the projects that are implemented are those that "look best on paper" but that become "the

Table 1. Construction cost and traffic performance of transport projects.

Source	Investment type	Construction cost: Sample	Construction cost: Ex post/ex ante cost (1)	Traffic: Sample	Traffic: Ex post/ex ante traffic (2)	Ratio of (2)/(1)
Flyvbjerg and Bester (2021) as well as Flyvbjerg (2016)	Bus rapid transport	6 projects	1.41	4 projects	0.42	0.30
	Rail	264	1.40	74	0.66	0.47
	Tunnels	48	1.36	23	0.81	0.60
	Bridges	49	1.32	26	0.96	0.73
	Roads	869	1.24	532	0.96	0.77
	Total[a]	1,603	1.39	786	0.94	0.68
Ansar et al. (2016)	Roads	74 projects in China	1.275	137 projects in China		
	Rail	21 projects in China	1.415	19 projects in China		
	Total	95 projects in China	1.306	156 projects in China	0.95	0.73
Flyvbjerg (2009)	Roads			183	1.095	
	Rail			25	0.514	
Næss et al. (2006)	Roads			15 metropolitan roads	1.063	
				55 non-metropolitan roads	1.1–1.2[b]	
				7 toll roads	0.939	
				63 non-toll roads	1.115	

[a]Ratios calculated as weighted averages.
[b]Includes 15 roads in disadvantaged region eligible for support from EU Structural Funds, and 40 roads in other non-metropolitan areas.

worst, or unfittest, projects in reality, in the sense that they are the very projects that will encounter most problems during construction and operations in terms of the largest cost overruns, benefit shortfalls, and risks of non-viability". Referring to the "average project", Flyvbjerg (2016, p. 182) mentions the simultaneous presence of cost overruns and benefit shortfalls, which are "bad for viability". Finally, Flyvbjerg and Bester (2021, p. 401) conclude: "Such systematic and significant bias in cost–benefit analysis is likely to lead to resource misallocation, including initiating investments that ultimately turn out to have negative net benefits and should never have been started".

In their study of the sample of 95 road and rail infrastructure projects in China, Ansar et al. (2016) go one step further and conclude: "Due to a propensity to cost overruns and benefit shortfalls, the typical infrastructure investment destroys economic value" (ibid, p. 377). They base this conclusion on the reported availability of data on economic costs and benefits for 65 of the projects, of which 55% had an ex post benefit–cost ratio below 1, and another 17% had a lower-than-predicted benefit–cost ratio.

Flyvbjerg and Bester (2021) point out that their results are " … as much about impact prediction as about cost–benefit analysis … " (p. 398), albeit without specifying what they mean by impact. The references above to resource allocation, economic welfare, and cost–benefit analysis imply that the analyses adopt a welfare-economic perspective, where the economic impact of a public policy intervention, like investment in transport infrastructure, is measured as the change in social welfare (the sum of changes in consumer, producer and other relevant surpluses) caused by that intervention. The magnitude of the change in social welfare is estimated by means of a CBA.

This, then, raises the question whether the data in the Flyvbjerg database can be used to approximate changes in social welfare. A recent article (Välilä, 2024) contrasts the scope and measurement of variables in the Flyvbjerg database against CBA, and it assesses to what extent the reported forecast errors can be used to draw conclusions about the social welfare consequences of projects in the database. Välilä (2024) concludes that the forecast errors in benefit–cost ratios based on construction costs and first-year traffic demand proxy forecast errors derived from a fully-fledged CBA only under unrealistically restrictive conditions. Furthermore, the simulations presented in Välilä (2024) on the basis of forecast errors reported by Flyvbjerg (2016) as well as Flyvbjerg and Bester (2021) suggest that those errors do not imply likely loss of social welfare, as large as they are.

As a consequence, while the Flyvbjerg database has generated unique analyses of systematic forecast errors and their causes in the construction costs and first-year demand for transport infrastructure projects, its hard data do not allow for any further-reaching assessment of those projects' economic impacts. To estimate the social welfare consequences of transport infrastructure projects conclusively, it is necessary to carry out ex post CBAs of them. The available systematic evidence of ex post CBAs is therefore reviewed next.

2.2. Surveys of project-level ex post CBAs

This sub-section reviews surveys of samples of ex post CBAs of transport infrastructure projects. It draws on the recent article by Wang and Levinson (2023), which identifies earlier surveys that sample at least 10 projects. Table 2 summarises the surveys identified by Wang and Levinson, amended with surveys published concomitantly or subsequently.

The focus here is exclusively on the results of ex post CBAs of transport infrastructure projects, as the change in social welfare caused by a project constitutes the most comprehensive and consistent microeconomic measure of individual projects' economic impact. That said, ex post evaluations use a variety of approaches and metrics. Useful methodological overviews of ex post evaluations of transport projects in different countries, including the role of CBA in them, include Bonnafous (2021, 2014), Florio et al. (2018), Meunier and Welde (2017); Meunier et al. (2016); Nicolaisen and Driscoll (2016), Fitzroy (2014), as well as Florio and Vignetti (2013).

A few recent articles have analysed ex post metrics other than change in social welfare in samples of projects. They have considered projects' cost performance (Welde & Dahl, 2021); cost and schedule performance (Welde & Engelbø, 2024); wider impacts at local or regional level (Holmen, 2022; Welde & Tveter, 2022); wider impacts on manufacturing, business creation and employment (Kanai et al., 2024); as well as cost and benefit components without a summary performance metric (Highways England, 2017). Many of these articles have challenged the notion that major transport infrastructure projects perform poorly according to these metrics.

The available evidence base of samples of ex post CBAs shown in Table 2 is dominated by roads and European countries. There has been an increase in recent years in the number of surveys as well as the number of projects analysed in each survey. The surveys by Chapman (2024a, 2024b) are notable in that they sample the entire population of road projects in a country completed during a long time period. Volden and Welde

Table 2. Ex post cost-benefit analyses of samples of transport infrastructure investment projects (sample size 10 or more).

Source	No. and type of projects	Measure and scope of impact	Main findings
Chapman (2024b)	98 road and 40 rail projects in Sweden, completed 1998–2022 (roads) or 2005–2022 (rail) but excluding year 2006 due to inconsistent cost recording	NPV ratio[a]	Of the road projects, 86% had NPV ratio > 0 at 3 or 5 years after opening. Of the rail projects, 63% had NPV ratio > 0 at completion or after 5 years
Chapman (2024a)	138 road/highway major schemes in England, completed 2001–2016	BCR	93% with BCR > 1, 69% with BCR > 2, calculated at 1 or 5 years after completion
Wang and Levinson (2023)	59 road projects in Asia and Oceania and supported by the ADB, constructed 2004–2018, evaluated 2004–2020	Detailed economic analysis with NPV, EIRR, BCR available for 23 projects	"Vast majority" of projects with ex post BCR > 1
Volden and Welde (2022)	12 roads, 6 railway projects in Norway, all several years into operation	Benefit-cost efficiency (CBA decision criterion)	Most road projects, especially in urban areas, performed "excellently", railway projects scored lower
Halámek et al. (2021)	144 regional road modernisation projects aimed at improving road safety in two regions of the Czech Republic, implemented 2008–2015	Impact of change in investment cost and incidence of traffic accidents on NPV	Ex post investment costs on average below ex ante (increase NPV), but ex post injuries and fatalities on average above expectations (reduce NPV). Ex and ex post NPV negative for 40% of sample; positive for 36%; changed from positive to negative for 14%; changed from negative to positive for 10%
de Jong et al. (2020)	10 major transport infrastructure projects (5 road, 2 rail, 3 urban transport) supported by the European Regional Development Fund and Cohesion Fund during 2000-2013, evaluated 5–10 years after opening	CBA results (BCR), including some externalities but not all wider economic impacts, score (1-5) reported	Nine out of ten project with an ex post BCR > 1, though benefits for about half of the projects "clearly" below ex ante expectation
Odeck and Kjerkreit (2019)	27 Norwegian road projects in service for at least 5 years	NPV	17 projects with NPV > 0, 10 with NPV < 0
Welde and Volden (2018)	12 road and rail projects in Norway, opening year 2005–2011 and at least 3 years of operation	BCR (NPV); ex post CBA less thorough than ex ante, yet all main effects considered	7 road projects with positive NPV, 1 road negative NPV, 4 rail projects negative NPV
Kelly et al. (2015)	10 large transport projects (6 roads, 4 rail) in 8 countries that benefitted from European Union's funding, projects opened 2001–2010	NPV, BCR, IRR	Ex post NPV > 0 for 9 projects (of which 1 project borderline)

Note: NPV = Net Present Value; BCR = Benefit-Cost Ratio; (E)IRR = (Economic) Internal Rate of Return.
[a]"A net present value ratio of 0.5 means that for every 100 kroner bet you get back this plus another 50 kroner in return, seen over the entire life of the project" (p. 4). Chapman reports that this somewhat unusual measure of economic profitability is used by the Swedish Transport Administration (Trafikverket).

(2022) as well as Welde and Volden (2018) discuss sample selection and conclude that while their samples are not random, they are likely to be representative. The sample analysed by Wang and Levinson (2023) was selected based on criteria concerning data availability; the same goes for most of the earlier surveys, too.

The ex post CBAs included in the surveys in Table 2 differ in terms of how long the projects have been in operation and which benefits have been considered in the analysis. They are also likely to differ in terms of modelling and parameter choices (social discount rate, use of shadow prices, consideration of a counterfactual scenario) that have not been explicitly discussed in all articles. As a result, their results may not be fully comparable.

With those caveats noted, the headline results reported in Table 2 do not support the notion that transport infrastructure projects typically reduce social welfare. Nearly all surveys conclude that most projects increase social welfare (NPV > 0, BCR > 1). This result is consistent, at least qualitatively speaking, with the simulations presented in Välilä (2024), which suggest that even with significant construction cost overruns and first-year traffic shortfalls, projects can still increase social welfare if their entire life cycle is considered.

3. Aggregate-level evidence

There are both qualitative literature surveys and quantitative meta-studies of the link between transport infrastructure (investment) and economic activity (growth). While the former are helpful in identifying sources of heterogeneity and biases in empirical analyses, the latter go one step further and yield quantitative meta-estimates of the link between transport infrastructure and growth that control for those sources of heterogeneity and correct for the biases. Therefore, the focus of this section is exclusively on the latter.

To identify quantitative meta-studies of the link between transport infrastructure (investment) and economic activity (growth), searches were conducted in Google Scholar, Scopus, and in Web of Science. To increase the sample of studies, searches were conducted using search words covering also public capital and investment, as well as infrastructure capital and investment, in addition to transport infrastructure (investment). The searches were limited to studies dated in year 2000 or later, as surveys and meta-studies themselves cover individual empirical studies published also before that time.

The meta-analyses considered here comprise those that either focus exclusively on the elasticity of output or some other measure of economic activity with respect to transport infrastructure, or that include an analysis of that elasticity alongside other elasticities, and that have been published in peer-reviewed journals since 2000. Six such meta-analyses were found: Nuñez-Serrano and Velázquez (2017); Holmgren and Merkel (2017); Elburz et al. (2017); Celbis et al. (2014); Bom and Ligthart (2014); as well as Melo et al. (2013). Relevant meta-studies not published in journals, and therefore excluded from this analysis, include Adame García et al. (2017) as well as Josheski (2008).

The elasticity of output or some other measure of economic activity can be estimated using a number of empirical approaches, as reviewed by Romp and de Haan (2007). In his survey of empirical studies on the link between infrastructure and economic growth in developing countries, Straub (2008) reports that the most common theoretical basis for the empirical model specification is some form of aggregate production function, representing half of the surveyed studies. In their global survey of public capital and growth, Romp and de Haan (2007, p. 9) state that earlier literature has "generally assumed that public capital forms an element in the aggregate production function". Focussing on transport infrastructure, Melo et al. (2013, p. 697) refer to the production function approach as "[t]he general empirical approach adopted in the literature studying the linkage between transport investment and economic output".

The meta-studies employ an empirical model that can be generically represented as follows (Nuñez-Serrano & Velázquez, 2017):

$$\hat{\theta}is = \theta_0 + \vartheta is = \tilde{\theta} + \sum_{k=1}^{K} \alpha_k Z_{ik} + \tilde{\vartheta} is \quad (1)$$

where the subscript i refers to a specific estimate of the parameter of interest, θ, in a specific study (subscript s). If there is one true value for that parameter within the population, θ_0, and the differences between individual estimates are only due to random variation then $\vartheta_i \sim N(0, \sigma_i^2)$. However, if the studies report different values of θ also due to systematic differences in their methodologies, then such systematic differences can be captured by independent meta-regressors, call them Z and say that there are K of them in number. In many cases, a meta-regressor is a dummy variable, assuming the value 1 for a specific characteristic (say, if the infrastructure variable considered refers to transport infrastructure only) and zero otherwise.

The aim of meta-studies is thus to consider the reported values of $\hat{\theta}_{is}$; identify the set of meta-regressors Z that account for any systematic differences between the methodologies of the studies; and to estimate $\tilde{\theta}$ based on (1).

Appendix 1 summarises the meta-regressors used in the published studies. The number of meta-regressors employed varies between 27 (Melo et al., 2013) and 39 (Elburz et al., 2017).

In addition to the sources of heterogeneity identified in Appendix 1, the reported values $\hat{\theta}_{is}$ may be affected by publication bias, as statistically significant estimates of the elasticity may be more likely to be published than estimates that are insignificantly different from zero. The presence of publication bias can be assessed by considering whether there is a relationship between the reported elasticity estimates and their standard errors. Specifically, Melo et al. (2013) run a regression with the absolute values of the elasticity estimates explained by their standard errors and study-specific fixed effects, finding that there is indeed a systematic bias to report statistically significant estimates, but that there is no bias in favour of reporting positive elasticities as opposed to negative ones. It appears that they do not consider publication bias in their meta-regression. On the other hand, Holmgren and Merkel (2017) run a regression with the ratio of the elasticity estimates to their standard errors explained by an intercept and the inverse of the standard errors. As they find a statistically significant intercept, they conclude that there is a systematic bias; moreover, they also conclude that there is a bias in favour of reporting positive estimates. Therefore, they choose to control for publication bias in their meta-regression.

Table 3 summarises the main features and qualitative findings of the six meta-studies. It should be noted that the terminology in Table 3 follows the terminology used in the studies themselves and no attempt has been made to interpret or harmonise it.

The main insights from Table 3 can be outlined as follows. Transport infrastructure has a positive and statistically significant impact on economic activity, with the long-run impact greater than the short-run impact. The magnitude of the impact of transport infrastructure is comparable to or smaller than that of aggregate public capital. Among

Table 3. Quantitative meta-studies published in academic journals since 2000 on the macro-economic impact of transport infrastructure.

Source	Sample: Size (geographical scope)	Sample: Dependent variable (estimation approaches considered)	Sample: Infrastructure variables	Meta-regression: Main findings relevant for present study
Nuñez-Serrano and Velázquez (2017)	145 studies published 1983–2011, 1928 estimates of elasticities (both developed and developing countries)	Value Added at national or sub-national level (Cobb-Douglas production function only)	(1) Total public capital (2) Productive public capital (health, education, housing and community services, energy installations, communication and transport infrastructure) (3) Transport infrastructure (roads, railways, air and maritime infrastructure)	(1) Short- and long-term output elasticity with respect to total public capital positive and highly significant. (2) Short-term elasticity for transport lower than for total public capital, albeit less significantly. (3) Elasticity declines as public capital stock grows. (4) Geographical disaggregation lowers elasticities, underlining the importance of network effects.
Holmgren and Merkel (2017)	78 studies published 1973-2016, 776 estimates of elasticities (both developed and developing countries)	Aggregate, regional or sectoral output (production function, VAR, cost function)	Aggregate public or infrastructure capital, different kinds of transport infrastructure (air, port, rail, road), highway (road) capital	(1) Elasticities in declining order of magnitude: aggregate public capital, road/aggregate infrastructure, air transport. All elasticities positive and statistically highly significant. (2) Manufacturing and construction output elasticity of air transport infrastructure lower than that of roads. The opposite is true for agriculture and services. (3) Service output elasticity of rail infrastructure higher than that of roads, other sectoral output elasticities insignificantly different between rail and roads. (4) Agricultural output elasticity of port infrastructure higher than that of roads, other sectoral output elasticities insignificantly different between ports and roads. (5) Regional level elasticities lower than national-level ones unless spillovers taken into account; hence, cross-regional spillovers exist.
Elburz et al. (2017)	42 studies published 1995–2014, 912 estimates of elasticities (EU countries, U.S.A., other countries)	Probability of positive, negative or insignificant elasticity of regional growth with respect to public infrastructure.	Public infrastructure, land transport, air transport, water transport, telecommunications, roads, railways, airports, ports, other infrastructure.	(1) Probability of a strongly significant positive elasticity higher for land transport infrastructure. telecommunications and roads. (2) Air transport, railways and port infrastructure tend to have a negative impact on regional growth.

				(3) Considering inter-regional spillover effects reduces the probability of a positive elasticity.
Bom and Ligthart (2014)	68 studies published 1983–2008, 578 estimates of elasticities (OECD countries only)	Private sector output, GDP, or Gross Regional/State Output (production function)	(1) Total public capital (2) Core infrastructure (transport, utilities)	(1) Short run output elasticity of public capital positive and highly significant. Long run elasticity higher than short-run. (2) Regional-level elasticities substantially smaller than national-level elasticities, so inter-regional spillovers exist. (3) Elasticity of core infrastructure higher than that of other types of public capital. (4) Output elasticity transport infrastructure not significantly different from total public capital.
Celbis et al. (2014)	36 studies published 1999–2008, 542 estimates of elasticities (both developed and developing countries)	External trade (export or import regressions)	Indicators of infrastructure assets or services, measured for individual types of infrastructure or as a composite index of several sectors, including transport.	(1) Elasticity of net trade with respect to own infrastructure 0.3. (2) Land transport infrastructure has bigger impact on trade than maritime, air, communication or composite infrastructure. (3) Own infrastructure more important for exports and imports of developing countries than developed countries. (4) Lower elasticities in studies at sub-regional or firm level; macro-level elasticities higher than micro-level due to positive spillover effects.
Melo et al. (2013)	33 studies published 1988–2012, 563 estimates of elasticities (Europe, US, other countries)	Regional, cross-regional or cross-national output (production function, VAR)	Monetary or physical measures of transport infrastructure, either aggregated or for individual modes of transport (road, rail, port/ferry, airport)	(1) Output elasticity of transport infrastructure positive and highly significant. (2) Output elasticity of road infrastructure higher than that of other transport modes. (3) National-level elasticities higher than regional-level ones, due to either spurious correlations or spatial spillovers. (4) Long-run elasticities higher than medium- or short-run elasticities.

Table 4. Meta-regression estimates of elasticities of output with respect to different types of infrastructure.

Source	Total public capital	Composite infrastr	Transport infrastr	Road	Rail	Air	Port/ferry
Nuñez-Serrano and Velázquez (2017)	<u>0.132</u>*** (short term), 0.161*** (long term)	0.146 (productive infrastructure)	0.11*				
Holmgren and Merkel (2017)	0.343***	<u>0.232</u>***		0.232***	0.189	0.165***	
Bom and Ligthart (2014)	<u>0.106</u>*** (short run), 0.145*** (long run)	0.153** (transport and utilities)	0.083				
Melo et al. (2013)			0.2077–0.2164***	0.0878**	<u>0.0277</u>*** (0.371)	<u>0.0277</u>*** (0.0273)	<u>0.0277</u>*** (0.682)

Note 1. Baseline estimate is <u>underlined</u>; other estimates, if statistically insignificant, are shown in parentheses and are insignificantly different from the baseline.
Note 2. ***, **, * indicate statistical significance at 1%, 5%, and 10% level, respectively. No star indicates no statistically significant difference to the baseline estimate.

different types of transport infrastructure, land transport and especially road infrastructure have the highest impact on economic activity, although air transport and port infrastructure are important for agriculture, and air transport and rail infrastructure are important for services. From a regional perspective, land transport and especially road infrastructure are important, while other types of transport infrastructure may even be harmful. The elasticity of national-level transport infrastructure is higher than that in the regions, which has been interpreted by most studies as evidence of positive network effects or spillover effects between regions.[2]

The main findings listed above are common to all meta-studies considered. There are, of course, other interesting results that are not highlighted by all studies (e.g. the decline of output elasticity as the capital stock grows; the direction of causality and the presence of endogeneity bias) or not explicitly addressed by them due to their design (e.g. the impact on output elasticity of the level of economic development, or the size of country or region). Some of factors like these have been discussed in qualitative surveys but not in quantitative meta-studies; however, they are not pursed here, their potential importance notwithstanding.

Finally, Table 4 reports the numerical elasticity point estimates and their significance from the meta-regressions, corresponding to $\tilde{\theta}$ in (1). Only output elasticities are included, thus excluding Elburz et al. (2017), who measure probabilities, and Celbis et al. (2014), who measure trade, not output elasticities.

The meta-estimates of the elasticity of output with respect to aggregate public capital vary between 0.106 and 0.343. Those with respect to a composite measure of infrastructure vary between 0.132 and 0.232. The meta-estimates of the elasticity with respect to transport infrastructure span the range [0.106, 0.2164]. The elasticity estimates for road infrastructure at between 0.0878 and 0.232 are significantly higher than those for other types of transport infrastructure.

Despite controlling for many sources of heterogeneity among the underlying studies, the results of the meta-analyses still display a range of meta-estimates. Of course, the set of mega-regressors varies between the meta-studies, and even within each meta-study a number of alternative specifications of the meta-regression are generally reported, so to some extent one would expect their results to vary, too. For example, the meta-regression in Melo et al. (2013) that yields sub-sector specific elasticity estimates only contains those sub-sector specific effects as meta-regressors, thus not controlling for any other sources of heterogeneity, while the specification yielding the highest estimate of the elasticity with respect to transport infrastructure controls for as many as 15 sources of heterogeneity.

All in all, the meta-estimates of the elasticity of economic activity with respect to transport infrastructure at the aggregate level (country or regional level) are consistently positive and statistically significant. The impact appears to be particularly clear in the long run and at the country (network) level. Conversely, one can infer that analyses focussing on the short run or on higher levels of geographical disaggregation are more likely to produce low, insignificant or even negative elasticity estimates.

4. Discussion

The discussion in this section focusses on the comparability of the methods and impact measures used in the project-level and aggregate-level analyses reviewed above. Against that background, the congruence of the results in sections 2 and 3 can then be assessed.

Starting with the comparability of the methods of analysis, the approaches reviewed in section 2.1 and in section 3 have both been labelled meta-analyses, but there are important differences between them. The Flyvbjerg database has been said to allow meta-analyses, because it combines data from primary and secondary sources, and because there are data control and selection procedures in place to ensure the comparability of the raw data (Flyvbjerg & Bester, 2021, Appendix). The analyses based on the Flyvbjerg database are, however, non-replicable, because the database is not accessible. In contrast, the meta-analyses at the aggregate level collect parameter estimates from published empirical studies and seek to find the true value of the parameter by replicable regression analyses that control for differences between the underlying studies.

The impact measures available in the Flyvbjerg database comprise forecast errors in construction costs and in first-year traffic demand. As discussed in section 2.1, the large magnitudes of those forecast errors have been used to draw conclusions about the broader economic impact, specifically about the negative social welfare consequences, of transport infrastructure projects. However, as also discussed in section 2.1, it has been argued that such forecast errors are insufficient to draw any social welfare conclusions about entire life cycles of projects. This, together with the non-replicability of the analysis, means that the Flyvbjerg database cannot offer any conclusive or even indicative evidence of the social welfare impact of transport infrastructure projects.

To gain reliable insights about the economic impact of transport infrastructure projects, a systematic analysis of ex post CBAs of such projects is necessary. While the evidence base remains limited, especially as regards its coverage in terms of transport sub-sectors and countries, it has been growing in recent years. The results of published surveys of samples of ex post CBAs point to a positive social welfare impact for most projects. Although this does not constitute conclusive evidence, it can arguably be considered as cautiously indicative of a predominantly positive impact.

Despite the diverging project-level welfare conclusions drawn from the Flyvbjerg database on the one hand and from the surveys of ex post CBAs on the other hand, it is noteworthy that both point to a better performance – according to their respective performance metrics – of road projects than rail projects. This raises the question of what accounts for this difference and, more broadly, which factors determine the observed forecast errors (cost overruns and first-year traffic demand) and welfare impacts? The available quantitative evidence is limited to reporting performance differences between transport sub-sectors. Qualitatively, both deliberate strategic misrepresentation and cognitive biases have been identified in earlier literature as possible determinants of project performance and differences therein. Flyvbjerg et al. (2002, 2003); Næss et al. (2006); as well as Flyvbjerg (2009) compare forecast errors in road, fixed link and rail projects and consider a number of possible explanations for them. They conclude that deliberate strategic misrepresentation by project planners and promoters is the single best explanation for forecast errors in general, and for the relatively poorer performance of rail projects in particular. Specifically, Næss et al. (2006) compare the incentives of road and rail planners in Europe during the 1950s–1990s and conclude that rail planners had stronger incentives than road planners to overestimate traffic to get projects approved and financed. Welde and Volden (2018) conclude that in the Norwegian context, the forecasts of the benefits of rail projects are subject to a stronger optimism bias than the forecasts of the benefits of road projects; cost estimates are unbiased in both cases. On the other hand, Chapman (2024b) points to the consistency between NPV estimates ex ante and at completion, and to their improvement during operation in both road and rail projects as suggesting that neither cognitive biases nor strategic misrepresentation have been present in his sample of Swedish projects. He emphasises, however, that more granular data would be needed to analyse the reported performance differences by, for example, geographical location and projects' complexity.

At the aggregate level, in contrast, the meta-studies find an unambiguously significant and positive output elasticity of transport infrastructure. It might seem tempting to conclude, then, that the aggregate-level evidence supports the indication of a positive impact at the project level. One should, however, not succumb to that temptation, both because aggregate output and social welfare are different measures of economic impact, and because there is no reason to expect their movements to be equal in magnitude or even point in the same direction.

Aggregate output (or Value Added or Gross Domestic/Regional Product) measures the size of an economy as the recorded monetary value of final goods and services exchanged on formal markets during a given time period, possibly augmented by an estimate of some types of non-market production (Laird & Byett, 2023). Output elasticity, the most common aggregate-level measure of the economic impact of transport infrastructure, measures the relative change in output following a change in transport infrastructure investment or assets, keeping other things equal. In contrast, CBA measures the change in social welfare during the life cycle of a transport infrastructure asset, considering both market and non-market effects; valuing them at shadow prices; and considering the incremental impact of the project above and beyond a counterfactual scenario of what would happen in its absence.

Therefore, aggregate output and changes in social welfare are very different measures of economic impact (Laird & Byett, 2023). The relationship between changes in aggregate output (national income) and social welfare (consumer surplus) caused by a transport infrastructure investment is spelled out by Mohring (1976; as quoted in Laird & Byett, 2023). The conditions under which the two are equal in magnitude are restrictive, so it is not possible to articulate any general rule linking them. The consideration of producer and government surpluses complicates the comparison of the two impacts further, as they affect social welfare directly but output only under some conditions. Furthermore, non-market effects like environmental or health externalities associated with the provision of transport services affect social welfare directly but output only indirectly. Consequently, not only the magnitudes of the output and social welfare impact of a transport infrastructure project may differ, even their signs may in theory differ, too.

The results from the project-level analyses are therefore not directly comparable with the results from the aggregate-level analyses. This absence of congruence has been recognised before (Vickerman, 2007b), and it has been attributed to the different purposes, metrics and methods of project-level and aggregate-level economic analyses. The former concerns the assessment of the economic viability of transport infrastructure investment projects, often with the normative objective of determining whether an investment should be undertaken or not. The latter concerns the understanding of aggregate-level relationships from a positive standpoint, without any immediate normative objectives.

Yet, the two levels of analysis are interconnected and can inform one another. Notably, the review of the aggregate-level meta-analyses in section 3 produced two headline results that have important implications for the analysis of project-level impacts of transport infrastructure.

First, the result from the aggregate-level analyses that the long-term impact of transport infrastructure tends to exceed its short-term impact confirms the view discussed in section 2 that a fully-fledged CBA covering the entire life cycle of a project is necessary to gauge its social welfare consequences. Shortcuts focussing on first-year demand as a proxy for life cycle benefits are inadequate and misleading.

Second, consider the result that the output elasticity of transport infrastructure is higher at the level of a country (or a network) than a region (or part of a network), which the aggregate-level literature has attributed to the presence of positive network or regional spillover effects. Network effects have an impact on social welfare, too, but that impact is "notoriously difficult to identify and model" (Vickerman, 2007a, p. 598). Regional spillover effects, in turn, reflect the presence of wider economic impacts beyond the user benefits and direct costs associated with a transport infrastructure project. Such wider impacts emerge outside the project's primary market in the presence of market failures or distortions. They include productivity effects associated with economic density and scale as well as competition; induced private investment over and above investment linked to direct user benefits; as well as labour market effects (Venables, 2016). Wider impacts may emerge in markets for final outputs – for instance increased competition leading to higher output – and therefore affect the measured output elasticity directly, or else they may affect output and output elasticity indirectly through markets for inputs or through non-market effects, such as knowledge spillovers among firms in a cluster. Wider impacts are often considered as benefits, but they can be negative, too, like some regions losing out to growing agglomerations of economic activity (Rothengatter, 2017; Vickerman, 2007b). Vickerman (2017) suggests that wider positive impacts of large-scale transport projects may increase benefits by 10-20% but that this estimate is gross of any wider negative impacts and dependent on the economic structure, level of transport costs, and on the impact of investment in transport infrastructure on both.

The inclusion of network effects or wider economic benefits and costs in the appraisal of large transport infrastructure projects is not entirely uncontroversial (Vickerman, 2007b, 2017). In principle, they should be estimated to complement a CBA of primary market welfare impacts, but in practice they are difficult to identify, estimate and value. Network effects or wider benefits are therefore a questionable justification for an otherwise marginal or unviable project (Vickerman, 2007a, 2017); besides, a transport project is the first-best solution only to a transport problem faced by users, while other instruments are likely to be more efficient in solving problems in other markets.

Nevertheless, the observed aggregate-level relationship between output elasticity and level of geographical aggregation provides evidence of the presence of network effects and wider impacts. That evidence lends support to scholars who have argued in favour of their systematic yet methodologically rigorous consideration in the appraisal and evaluation of major transport infrastructure projects (Rothengatter, 2017; Venables, 2016; Vickerman, 2007a, 2007b, 2017). The challenge herein lies in the development of methods and models to estimate wider impacts that are cost-effective and whose output is consistent with and can be integrated into the CBA results of a project's social welfare consequences on its primary market.

5. Conclusions

Project-level meta-studies based on the Flyvbjerg database conclude that transport infrastructure investment projects suffer from systematic construction cost overruns and first-year traffic demand shortfalls and, as a corollary, that they are likely to reduce social welfare as a rule. While the conclusions concerning systematic and large forecast errors in construction costs and first-year demand may well be accurate, data on those forecast errors is insufficient to draw conclusions about transport infrastructure projects' social welfare consequences over their entire life cycles. To that end, systematic evidence from ex post CBAs is necessary. A number of recent studies have surveyed samples of ex post CBAs of transport infrastructure projects, challenging the notion that they destroy social welfare.

The aggregate-level meta-studies provide robust evidence of a positive and statistically significant relationship between transport infrastructure and economic activity, and of the fact that that relationship is greater in the long run and at higher levels of geographical aggregation. There are possible concerns about publication bias, which has been accounted for in some meta-studies but not in others, and about the sensitivity of the results to the set of meta-regressors employed. But those issues concern primarily the accuracy of the estimated output elasticity, less so its sign and statistical significance.

Consequently, our understanding of the project-level economic impact of transport infrastructure is improving but remains inconclusive, while we have strong evidence of a significant and positive impact at the aggregate level. This conclusion is directly opposite to that of Ansar et al. (2016), discussed in the introduction, which suggested that the project-level evidence is more compelling than the aggregate-level evidence.

Another conclusion from the review of evidence in sections 2 and 3 above is that there is no paradox between the project-level and aggregate-level results, as suggested by Ansar et al. The project-level analyses and the aggregate-level analyses have different objectives, measures of economic impact, and methods of analysis. There is no reason to expect that their results be congruent; hence, there is no basis for suggesting that the results constitute a paradox.

Despite all these differences between the project-level and aggregate-level analyses, they can still provide valuable insights for one another. The aggregate-level meta-analyses identify the time horizon and the level of geographical aggregation as significant determinants of the economic impact of transport infrastructure. Project-level analyses based on the Flyvbjerg database have, for reasons of data availability, focussed on first year traffic demand as a proxy for life-cycle benefits. The aggregate-level finding concerning higher long-term impact calls this approach into question and supports the view that a fully-fledged ex post CBA is necessary to estimate life-cycle impacts. Similarly, project-level analyses focus often on estimating primary market welfare impacts and ignore network effects and wider impacts, for a variety of good reasons. Yet, the aggregate-level finding concerning higher impacts at higher geographical levels of aggregation suggests that network effects and wider impacts – benefits but possibly also costs – are a typical, if not universal, feature of transport infrastructure. This is no novel insight among experts of cost–benefit analysis of major transport infrastructure projects, but it constitutes evidence in support of the consideration of network effects and wider impacts in project appraisal and evaluation, notwithstanding the well-understood challenges to that end.

The main priorities for future research are quite clear. First and foremost, additional ex post evidence of the social welfare consequences from large and preferably random samples of transport infrastructure investment projects would be highly valuable. There are well-understood reasons for the paucity of such evidence, but the publication of several recent surveys gives hope for more to come. Second, the incorporation of network effects and wider economic impacts into both ex ante and ex post CBAs of transport infrastructure projects remains controversial and difficult in practice, but the results of the aggregate-level meta-analyses underline the importance of developing rigorous but cost-effective ways to do just that, especially for major projects. Third, further analysis of the relationship between forecast errors (cost overruns and demand shortfalls) as well as projects' social welfare consequences would be valuable, including also a systematic analysis of how the determinants of forecast errors and economic impacts may differ. At the aggregate level, only two meta-studies have attempted to estimate the output elasticities of specific sub-types of transport infrastructure. Their results are quite different, so further work would be welcome to understand how and why output elasticities vary across transport sub-sectors.

Notes

1. The Flyvbjerg database appears to be related to the Oxford Global Projects database, which is used for commercial purposes. It says on its website: "We have the largest high-quality datasets on project performance in the world". That database reports that it contains data on 11,123 projects, of which 2,299 in transport. Web-link to a description of the Oxford Global

Projects database (accessed on December 27, 2024): https://static1.squarespace.com/static/5c312863266c07d084fcd39e/t/5c59e4aaec212d5cdf4d6b66/1549395166162/OxfordGlobalProjects.pdf.

2. In aggregate-level literature, network effects refer to the impact of one region's transport infrastructure on the productivity of transport infrastructure in other regions. Spillover effects refer to the impact of one region's transport infrastructure on the output of other regions. This distinction is clear in theory but difficult to make empirically (Di Giacinto et al., 2012).

Acknowledgements

I thank two anonymous reviewers, as well as seminar participants at Harvard's Growth Lab and The World Bank's InfraXchange for helpful comments and discussions. I also thank D'Maris Coffman for suggesting this topic for a lecture, and I am grateful to Juliano Denicol, Antonio Estache, José Doramas Jorge Calderon as well as Morten Welde for their comments on earlier drafts. All remaining errors are mine alone.

Disclosure statement

The author is employed by the European Investment Bank. The opinions expressed are personal and may not necessarily reflect the European Investment Bank Group position.

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Appendix 1. Sources of heterogeneity in primary studies and choice of meta-regressors in meta-studies

The left-most column identifies the fundamental source of possible heterogeneity among the elasticity estimates; the second column breaks that fundamental source down to distinct sub-categories; and the third and fourth columns list the specific meta-regressors that have been employed. The latter have been presented in two separate columns, as some studies use more aggregated measures (column three) than others (column four), or as a generic approach (column three) can be implemented in a number of separate ways (column four).

Source of heterogeneity	Dimension of heterogeneity	Meta-regressor	Meta-regressor
Measurement of capital	Scope of measurement	Total public capital	
		Core/productive infrastructure (transport, utilities)	
		Transport infrastructure	Road
			Rail
			Air transport
			Port
	Unit of measurement	Physical units	
		Composite index	
		Monetary value	Investment flow
			Capital stock
	Level of measurement	National	
		Sub-national	
Measurement of output	Scope of measurement	Value Added (GDP)	Total
			Sectoral
			Sub-national
		Private sector output	
	Level of measurement	National	
		Sub-national	
		Industry level	
		Firm level	
Empirical strategy	Approach	Production function	Infrastructure as an input
			Infrastructure in technology function
			Returns to scale
			Frontier function
			Spatial heterogeneity and dependence
			Controls
		Growth (or trade) regression	
		Cost function	

(*Continued*)

Continued.

Source of heterogeneity	Dimension of heterogeneity	Meta-regressor	Meta-regressor
		VAR	
	Time horizon	Short term	
		Medium term	
		Long term	
Data	Sample structure	Panel	Time trends addressed
		Time series	Time trends addressed
		Cross-section	
	Level of aggregation	National	
		Sub-national	
		Firm level	
	Sample period	Middle (median) year	
Estimation	Method	Time-series methods	
		Ordinary Least Squares	Unobserved effects (fixed or random)
		Instrumental variables	
	Controls	Level of national income	Developed/developing (low income)
			GDP (per capita)
		Geography	Region (e.g. EU)
			Country (e.g. U.S.A.)
		Level of government in charge of infrastructure	Central
			Sub-national
		Urbanisation	
		Congestion	
		Capacity utilisation/ cycle	
		Time effects	
Publication	Journal	Ranking of journal	
	Year		

Sources: Nuñez-Serrano and Velázquez (2017) Table 2 (pp. 326–327); Holmgren and Merkel (2017) Table 2 (page 17); Elburz et al. (2017) Table 2 (page 3); Bom and Ligthart (2014) Table 1 (pp. 900–901); Melo et al. (2013) Table 3 (p. 700); Nijkamp and Poot (2004) Table 3 (pp.102–103).